From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work

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Standard operating procedures (SOPs) in pharmaceutical manufacturing environments are legally binding documents, but their practical force depends on whether they are reachable at the moment action is required. This paper argues that SOP accessibility is not only a retrieval problem but an HCI problem about how access conditions shape interpretation upstream of formal task execution. Interface structure determines which procedures become visible and comparable — making retrieval design a tractable target for HCI intervention even within the validated-archive constraint. In GMP-regulated environments, practitioners often work through validated archives that cannot be modified, meaning that retrieval support must operate externally rather than by restructuring source documents. I develop and evaluate an external keyword-searchable retrieval layer over a corpus of 500 synthetic GMP SOPs and show that reducing retrieval burden does more than save time: it expands the procedural landscape practitioners consider available, supportable, and actionable. Findings from a think-aloud pilot study with $N = 20$ GMP-experienced participants show that keyword-based access reduced exploratory retrieval time from approximately six minutes to approximately one minute, that result cues were rated useful by 18 of 20 participants, and that all 20 indicated willingness to reuse the system in live archive conditions. Participants also described a broader shift: when procedures become easier to reach, they become easier to compare, easier to justify using, and less dependent on informal colleague mediation. I further introduce an AI support layer as a conceptual design extension addressing interpretive burden beyond retrieval by explaining why a result appeared, surfacing related procedures, and keeping alternative interpretations visible without resolving them. The paper makes three contributions: it introduces SOP accessibility as a design problem specific to GMP-regulated environments; identifies the validated-archive constraint as a novel design boundary for retrieval research; and argues that access conditions shape interpretive practice upstream of any formal task error, making retrieval design a question of accountability conditions, not only usability — and that judgment-preserving AI support can extend this intervention into the interpretive space that retrieval alone cannot reach.

Additional Key Words and Phrases: SOP accessibility, document retrieval, pharmaceutical manufacturing, GMP, interpretive variability, invisible work, regulated environments, judgment-preserving design

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1 Introduction

Standard operating procedures (SOPs) are among the most consequential documents in regulated manufacturing. In pharmaceutical production environments governed by Good Manufacturing Practice (GMP) regulations, SOPs define how processes must be performed, what counts as acceptable deviation, and what actions can be defended after the fact. Yet the practical challenge of consulting an SOP at the moment it is needed has received little attention in the HCI literature. Prior work on SOP non-compliance has focused on training gaps, motivational barriers, and procedure

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design rather than on the interaction conditions under which practitioners access documents in the first place [4, 8]. Krockenberger et al. [8] found that 18.4% of clinical trial coordination staff sought a colleague's guidance rather than retrieving the relevant SOP, concluding that the primary barrier was inadequate retrieval system design. That diagnosis has shaped subsequent system development in regulated environments, yet the design response it implies has not been developed for GMP-specific contexts, where the constraint of validated archives uniquely restricts what retrieval interventions are permissible.

This gap matters because access conditions shape interpretation. When practitioners cannot locate a relevant procedure quickly because archives are stored as scanned PDFs that block keyword search, because titles are too similar to distinguish, or because time pressure makes trial-and-error retrieval costly, they do not simply fail to comply. They navigate under constrained visibility. They work from memory, defer action, or ask colleagues. These substitutions occur before any formal task error and before any later assignment of responsibility. In this sense, the design of retrieval is not downstream support for already-defined work. It helps determine what procedural possibilities become reachable, comparable, and actionable in the first place.

I argue that document accessibility is an upstream condition on interpretation in regulated work environments. Retrieval burden shapes the range of procedures a practitioner encounters, compares, and can act on with confidence. This argument extends Suchman's (1987) account of situated action to the domain of procedural document use: the formal existence of a procedure does not determine its practical effect; what matters is what is visible and reachable at the moment action is required. In GMP archives, this question becomes a distinct HCI problem because validated source files cannot be modified. Interface structure determines which procedures practitioners can reach and compare — and when that structure fails to externalize the archive's logic, the burden of retrieval falls entirely on the practitioner rather than on the system. Any retrieval improvement must therefore operate externally, over metadata and interaction structure, without altering the documents themselves.

This paper reports a prototype-based pilot study in a pharmaceutical GMP environment. I developed an external retrieval layer — a keyword-searchable metadata index — over a corpus of approximately 500 procedural documents and evaluated it with $N = 20$ frequent SOP users using a think-aloud protocol. The prototype served simultaneously as a design intervention and as an analytic probe: by observing how participants navigated the system, I was able to make upstream retrieval burden visible in a way that conventional error analysis cannot. Findings show that reducing retrieval effort shifted participant attention from document navigation to document interpretation, and that the friction revealed was structural rather than individual, traceable to archive design rather than user failure. A conceptual extension — an AI support layer that addresses interpretive burden beyond retrieval — is introduced as a design direction grounded in a recognized limitation of the pilot prototype.

The contributions of this work are threefold. First, I introduce SOP accessibility as a design problem within GMP-regulated environments and identify the validated-archive constraint as a novel design boundary for retrieval research. Second, I show that retrieval design changes more than navigational efficiency: it changes the interpretive conditions under which practitioners decide what procedures are available to use and justify. Third, I propose an AI support layer as a judgment-preserving design extension — one that expands interpretive cues without automating compliance decisions — and identify the design principles that distinguish it from conventional AI retrieval support.

2 Related Work

SOP Non-Use and the Limits of Procedural Infrastructure

Prior research has documented that practitioners routinely fail to consult standard operating procedures at the point of action. Krockenberger et al. [8] found that 18.4% of clinical trial coordination staff sought a colleague's guidance rather than retrieving the relevant SOP, attributing this pattern to inadequate retrieval system design rather than individual non-compliance. A broader literature on procedural adherence confirms that non-compliance with written procedures is rarely a matter of individual negligence; rather, it reflects structural conditions including inaccessible documentation, outdated archives, and the gap between procedures as designed and tasks as performed [4]. Yet both lines of work leave a critical question unaddressed: even when retrieval is improved, does better access simply make documents easier to find, or does it reshape the interpretive work that follows? The present study takes up this gap, examining not only whether practitioners can locate SOPs, but how variability in access conditions shapes what they can compare, justify, and act on.

Professional Document Work and the Persistence of Retrieval Difficulty

The difficulty of locating authoritative documents is not unique to pharmaceutical manufacturing. Sellen and Harper [11] demonstrate that digital transitions do not resolve the retrieval problems of document-intensive work; rather, they shift the role of documents without eliminating the navigational demands that practitioners face. Harper's (1998) ethnographic study of documentation practices at the IMF similarly shows that documents do not operate as neutral carriers of information: their accessibility structures the work that surrounds them, and gaps in retrieval support create invisible friction that is rarely captured in organizational accounts. Russell-Rose et al. [10] extend this argument across healthcare, legal, and patent work contexts, demonstrating that retrieval difficulty is a structural feature of professional document environments, not a correctable surface problem. The present study examines how this structural difficulty manifests in GMP contexts, where the constraint of validated archives uniquely restricts what retrieval interventions are permissible.

Situated Action and the Inaccessible Procedure

The theoretical basis for treating access conditions as consequential for interpretation comes from Suchman's (1987) account of situated action. Suchman argues that human action is not plan-driven but shaped by the immediate conditions of its execution — what is visible, reachable, and actionable in the moment of practice. Applied to SOP use, this framework implies that a procedure's formal existence does not determine its practical effect; what matters is what is accessible to the practitioner at the moment action is required. Unstable or inconsistent access to SOPs thus becomes not a logistical inconvenience but a structural source of interpretive variability, and therefore a design condition on judgment rather than a mere documentation problem.

Invisible Work and Upstream Labor

When SOPs are inaccessible, practitioners do not simply fail to act; they perform substitutive work — asking colleagues, reconstructing procedures from memory, or deferring action. Star and Strauss [12] show that work of this kind tends to become invisible: it is not captured by formal indicators, not measured by quality systems, and not attributed to the access failure that produced it. In regulated manufacturing, this invisibility has direct implications: the retrieval burden created by inaccessible documentation sits upstream of any formal error record, and is therefore structurally absent

from CAPA analysis and audit review. The prototype-based study reported here makes this upstream burden legible — treating it not as individual user difficulty but as a structural condition of the archive that retrieval interventions can address without modifying validated source files.

Think-Aloud Methods and Prototype-Based Evaluation

The methodological approach in this study draws on an established body of HCI research on think-aloud evaluation and prototype-based inquiry. Ericsson and Simon [2] provide the theoretical foundation for think-aloud protocols as a data collection method: concurrent verbalization surfaces the cognitive processes underlying task performance without substantially altering their course, making it a particularly appropriate method for studying navigation and search behavior under realistic task conditions. Nielsen [9] operationalizes this approach within usability engineering, demonstrating that think-aloud studies with relatively small participant samples reliably surface the majority of usability problems in an interface; subsequent work confirms that $N = 20$ expert participants in domain-specific populations reliably surfaces the dominant usability and friction patterns [9]. Jaspers et al. [7] apply this method in a regulated clinical context directly analogous to the present study: using think-aloud protocols and video recording to analyze how clinicians searched through paper-based patient records, they developed a prototype interface that reduced retrieval burden without altering source documents — a design constraint that mirrors the validated-file requirement in pharmaceutical GMP environments. Hutchinson et al. [6] further establish the logic of using a prototype as both an intervention and an analytic probe: a technology probe is simultaneously a design artifact and a research instrument, making it possible to study how access conditions shape behavior rather than simply to evaluate whether a given interface is usable.

AI-Assisted Retrieval and Judgment-Preserving Design

Within HCI, work on AI-assisted decision making has established that interface structure actively shapes which information becomes salient to practitioners navigating complex document environments. Wang and Yin [14] demonstrate that explanation interfaces alter how decision makers engage with outputs depending on domain expertise — findings that establish retrieval interfaces as active shapers of which procedures become visible and comparable rather than neutral conduits. Green and Chen [3] show that algorithm-in-the-loop settings systematically constrain the conditions under which independent judgment is possible, grounding the judgment-preservation constraint's design logic in this study: when retrieval structure determines which procedures practitioners can reach and compare, external layers that preserve interpretive authority are not merely a design preference but a structural requirement of regulated environments.

Prior work has not examined SOP accessibility as a design problem within pharmaceutical GMP environments, where document integrity constraints uniquely restrict retrieval system design. This study addresses that gap directly.

3 Method

Study Context

This study was conducted in the quality control department of a pharmaceutical manufacturing facility operating under 21 CFR Part 210/211 Good Manufacturing Practice regulations. The procedural archive used in this study consisted of 500 synthetic quality control SOPs generated programmatically in Python. Because actual facility SOPs are subject to non-disclosure obligations, the corpus was constructed to reflect the structural and terminological characteristics of

real GMP quality control documentation — including document identifiers, process category conventions, procedural language, and keyword distributions typical of QC operations — without reproducing proprietary content. Documents were rendered as PDFs and parsed to extract metadata for indexing. This approach preserves the ecological validity of the retrieval problem while maintaining confidentiality requirements. This construction approach reflects a domain constraint rather than a methodological preference: GMP procedural documents are proprietary quality records protected under confidentiality obligations that preclude their use in external research without facility authorization. Synthetic generation from structural parameters is the standard approach for research in regulated manufacturing contexts where live data access is unavailable. The validated-file constraint that defines the core design boundary of this work applies equally to synthetic and live corpora in GMP environments.

Phase 1: Metadata Mapping and Keyword Analysis

I conducted a systematic analysis of the full procedural archive to construct a searchable metadata index. For each document, I extracted the document identifier, title, process category, and keywords derived from the document content. Where keyword coverage was sparse, I supplemented extraction with manual annotation based on process area and task type. I also mapped structural overlap across the corpus: documents sharing similar titles, overlapping keyword sets, or belonging to the same process category were flagged as high-ambiguity clusters. This analysis identified the specific points in the archive where practitioners faced the greatest navigational difficulty — not from individual document complexity, but from the structural similarity of adjacent procedures.

Phase 2: Prototype Development

Based on the metadata index, I developed an external retrieval layer as a Streamlit web application (Figures 1 and 2). The prototype allows practitioners to enter task-relevant keywords — terms describing what they are trying to do, rather than document titles or identifiers — and returns a ranked list of relevant SOPs with document identifiers, process categories, and contextual snippets. Match strength is indicated at three levels (Strong: 5+ matches; Good: 2–4 matches; Weak: 1 match) to support result discrimination before opening documents. The system operates entirely over the metadata index; it does not modify, version, or interact with the validated source files in any way. Human interpretive authority is preserved: the system surfaces candidate documents, but practitioners retain full responsibility for reading, applying, and judging the procedures returned.

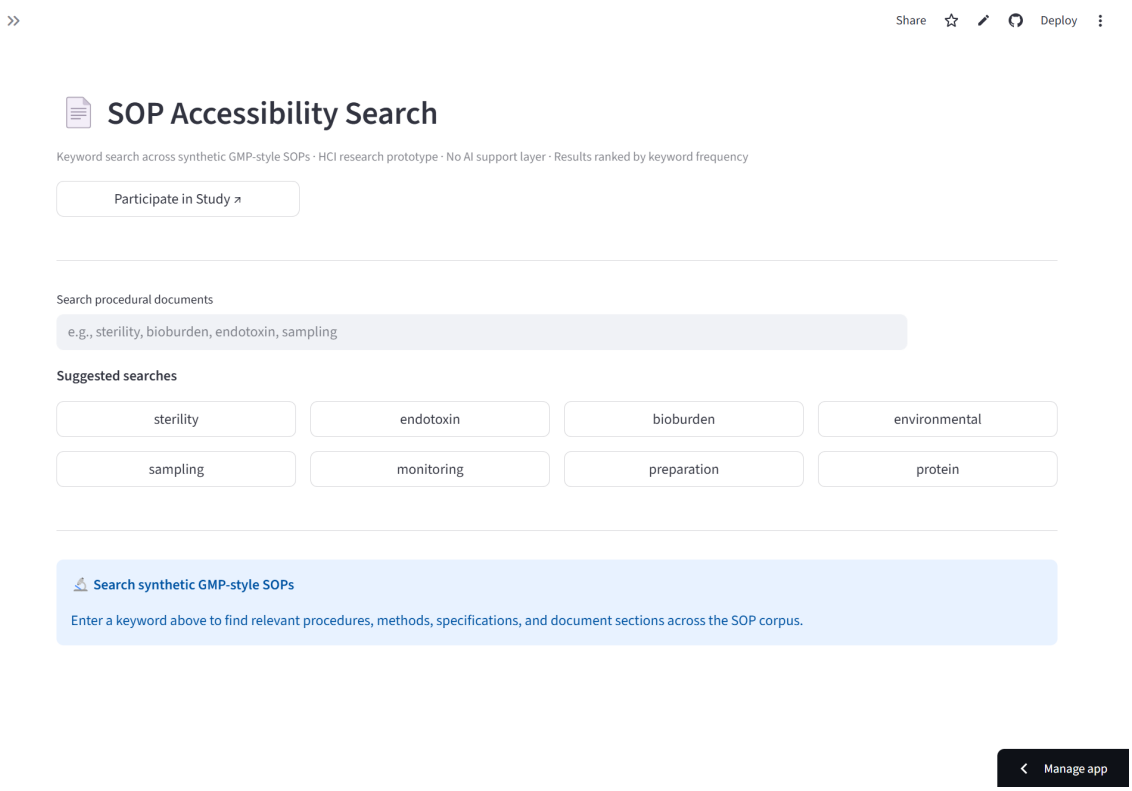


Fig. 1. The SOP Accessibility Search prototype at rest, showing the keyword input field and suggested search terms drawn from the synthetic GMP corpus. The interface decouples retrieval from archive naming conventions, allowing practitioners to enter task-relevant terms rather than document identifiers or titles.

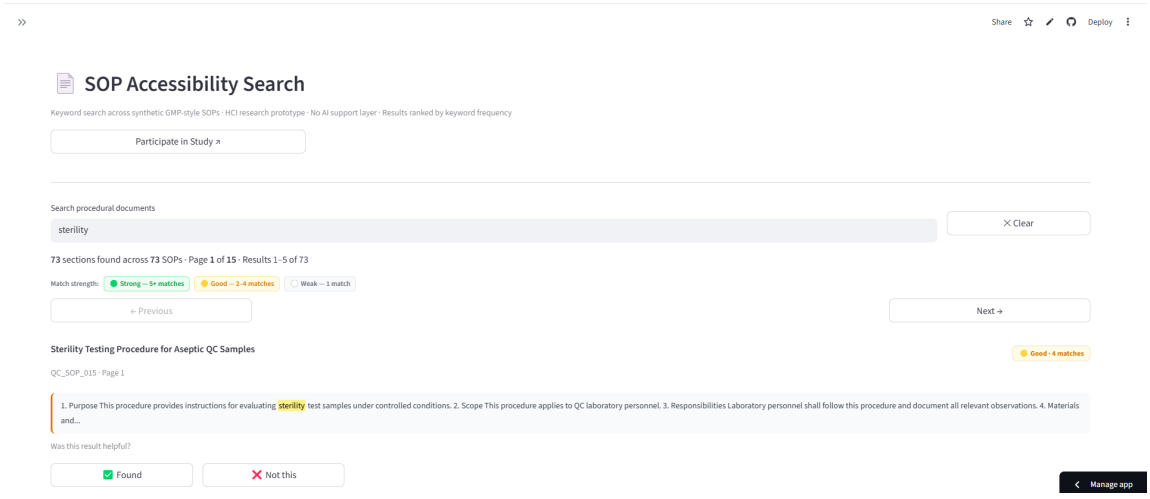


Fig. 2. Search results for the query "sterility," returning 73 sections across 73 SOPs. Results are ranked by match strength (Strong / Good / Weak) and include document identifiers, section locations, and contextual snippets with matched terms highlighted.

Phase 3: Pilot Evaluation

I evaluated the prototype with $N = 20$ participants recruited from the same GMP quality control team. All participants were frequent SOP users with daily procedural document search experience. This sample size is consistent with established think-aloud methodology, in which $N = 20$ expert participants in a domain-specific population reliably surfaces dominant usability and friction patterns [9]. Recruiting from a single expert team was a deliberate methodological choice to ensure consistent baseline familiarity with GMP retrieval conventions, prioritizing internal consistency over sample diversity. This design choice may have inflated usability and acceptance ratings relative to a more diverse sample — a limitation acknowledged below. Participants were asked to complete a series of realistic search tasks using the prototype — tasks drawn from their own reported search scenarios — while thinking aloud throughout. Sessions were conducted individually; a researcher observed and took field notes without intervening in participant navigation. Approximate retrieval times were recorded informally as a descriptive measure of navigational effort; these are exploratory observations rather than controlled measurements.

Following the search tasks, participants completed a brief post-task questionnaire assessing ease of finding relevant sections, usefulness of result cues, overall intuitiveness, and willingness to reuse the system in similar work contexts. Open-ended prompts invited participants to describe remaining friction points and to articulate how the prototype differed from their current search practice.

Think-aloud data were analyzed following Ericsson and Simon's (1993) protocol analysis approach, with particular attention to moments of hesitation, uncertainty, and navigational substitution — instances where participants paused, reworded queries, or expressed doubt about whether a returned result was the correct procedure.

4 Findings

Access Burden as an Interpretive Condition

Participants consistently described SOP retrieval as a meaningful burden in their daily work — not a minor inconvenience but a recurring cost that competed with task execution. One participant described the experience directly: "dying of busyness while SOP search is also work." A second participant framed the burden in terms of its downstream effect on judgment: "when you spend time just finding the right SOP, you are already tired before you even start reading it." These accounts position retrieval not as a neutral precondition for work but as part of the interpretive conditions under which work begins. Before a practitioner reads, compares, or decides, they have already spent effort navigating whether the relevant procedure is even reachable.

The prototype reduced this burden in observed use. Exploratory retrieval times dropped from approximately six minutes under current conditions to approximately one minute with the prototype, though these are informal observations rather than controlled measurements. Participants reported that keyword-based access allowed them to enter task terms rather than document titles or identifiers, reducing reliance on archive naming familiarity. Result cues — document identifiers, category labels, match strength indicators, and contextual snippets — were rated as useful (4.5/5) in helping narrow the search space before opening documents. 18 of 20 participants rated the system as intuitive overall, with the 2 exceptions attributing their hesitation to result discrimination difficulty in the synthetic corpus rather than interface design. All 20 participants indicated willingness to reuse the system in live archive conditions.

Accessibility Constraints on Behavioral Repertoire

The most consequential finding concerned the scope of action that practitioners allow themselves. Participants described a pattern in which the practical difficulty of locating unfamiliar SOPs leads them to operate within a personally known subset of the archive — consulting procedures they have used before rather than searching for procedures that might be more directly applicable. As one participant put it: "when you cannot find a SOP easily, you end up acting within the SOPs you already know." In GMP environments, this has direct regulatory implications. SOPs function as the formal basis for action: they define what practitioners are authorized to do and provide the documented justification for any action taken. When accessibility constraints limit the set of procedures a practitioner encounters, they also limit the range of authorized actions available to them — not because the procedures do not exist, but because they are not reachable.

The prototype made a wider procedural landscape accessible, and participants reported that this changed more than navigational ease. It changed what they felt able to consider, compare, and defend using. In this sense, accessibility interventions do not merely accelerate a pre-existing workflow. They reshape the behavioral repertoire within which interpretation occurs.

This finding directly extends Krockenberger et al.'s (2008) observation that 18.4% of practitioners substitute colleague consultation for SOP retrieval: the present study shows that even among practitioners who do retrieve SOPs, retrieval difficulty constrains which SOPs they retrieve — producing a second, subtler form of behavioral restriction that prior work has not examined.

Nomenclature Ambiguity and Onboarding Labor

A second finding concerned the distribution of retrieval burden across experience levels. Senior practitioners described a recurring onboarding burden: new staff, unfamiliar with equipment naming conventions and archive organization,

routinely struggle to distinguish between similarly named procedures. A representative example: a senior practitioner directing a new hire to consult the UPLC SOP found that the new hire searched for the HPLC SOP instead, not recognizing the distinction between the two instrument types. Under current archive conditions, resolving this kind of ambiguity requires intervention from a more experienced colleague — creating a pattern of informal knowledge transfer that is not captured in any formal quality metric.

The prototype partially addressed this. Keyword-based retrieval reduced dependence on exact nomenclature, and result snippets provided contextual differentiation to help less experienced users distinguish between adjacent procedures. Participants anticipated that this could reduce the informal SOP-guidance labor that senior practitioners currently provide — labor that, consistent with Star and Strauss [12], is invisible to the quality system and unmeasured by any existing indicator.

This matters analytically because it shows that accessibility burden is not evenly distributed. Experienced practitioners carry less of it because they have already internalized the archive's naming logic, while less experienced practitioners bear more of it precisely because the system does not externalize that logic for them. The archive therefore redistributes interpretive burden by experience level.

Remaining Friction

Not all friction was resolved. Participants noted two categories of residual difficulty. First, no-result states remained ambiguous: when a search returned nothing, it was unclear whether the issue was a wording mismatch or a genuine gap in archive coverage. A partial design response is to surface empty results as signal rather than failure — indicating that the absence of a result is itself information about the archive structure, not evidence of user error. The AI support layer introduced in Section 5 addresses this directly through an "empty results as signal" design principle.

Second, 2 of 20 participants reported difficulty discriminating between results when multiple similar procedures appeared in the result set. Both attributed this to the synthetic corpus's terminological regularity. This observation suggests that discrimination may be easier in production conditions with messier naming conventions, while also confirming that terminological overlap is a structural feature of GMP archives that any retrieval layer must address.

5 Design Extension: AI Support for Interpretive Burden

The retrieval prototype addressed navigational burden — the cost of locating a relevant procedure. It did not fully address what follows: the interpretive burden of determining which retrieved procedure is most applicable, how its scope relates to adjacent procedures, and where ambiguity in its language may produce inconsistent application. Participants who successfully located relevant SOPs still reported uncertainty about whether they had found the *right* one, and about how to handle cases where two plausible procedures overlapped in scope.

To address this residual interpretive burden, I developed a conceptual extension: an AI support layer that expands the set of retrievable and comparable procedural cues, making ambiguity more visible without replacing human interpretation (Figures 3 and 4). This layer has not been subjected to a separate empirical evaluation; it is presented here as a design direction grounded in the limitations identified by the pilot study, and as an instantiation of the judgment-preservation principle that governs the retrieval prototype.

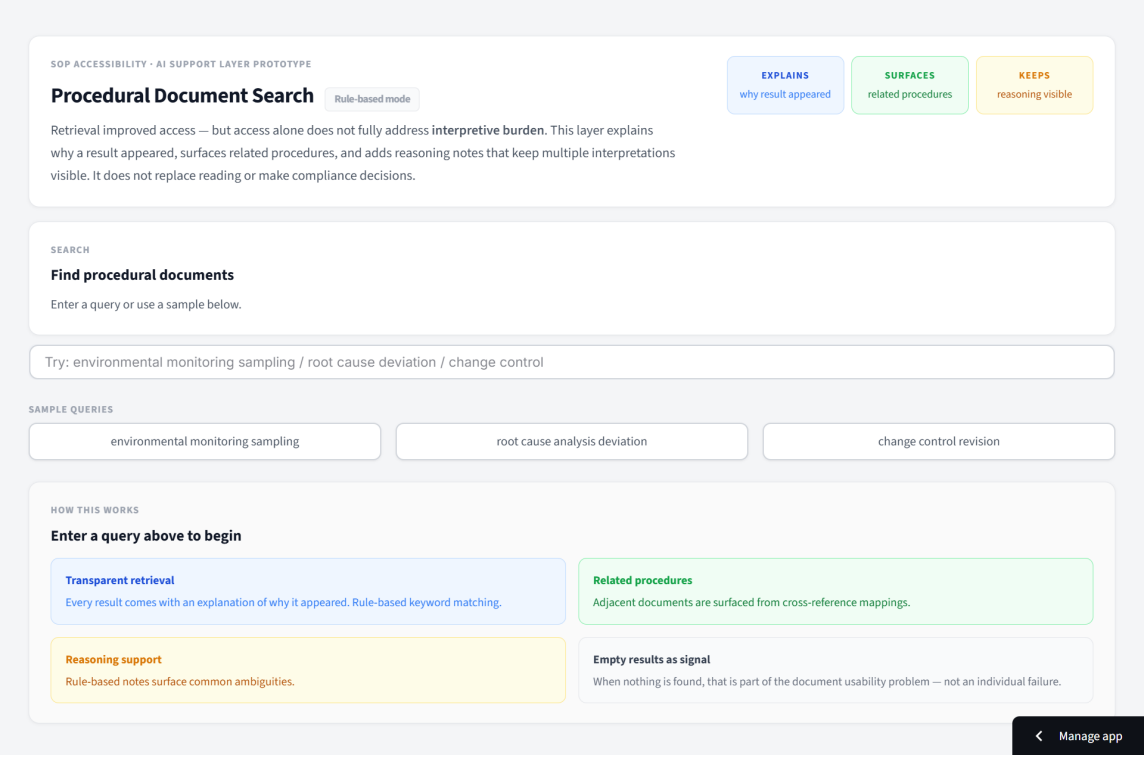


Fig. 3. The AI support layer prototype at rest, showing three design principles made explicit in the interface header: EXPLAINS (why a result appeared), SURFACES (related procedures via cross-reference mapping), and KEEPS (reasoning visible by surfacing common ambiguities).

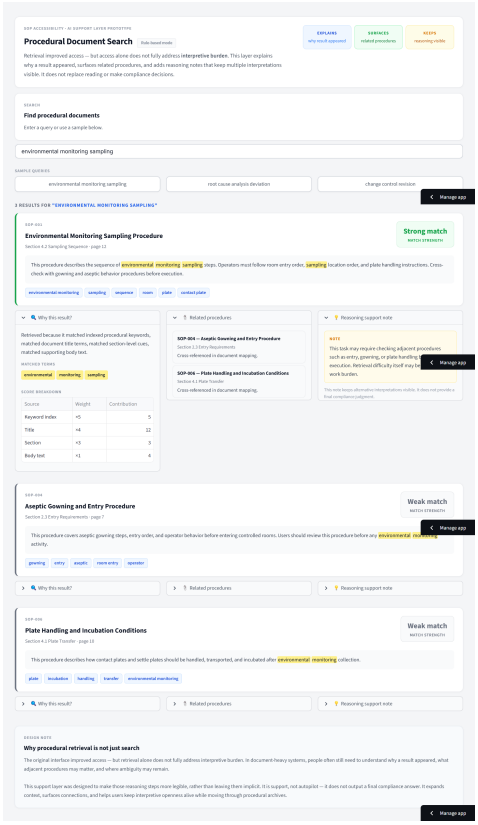


Fig. 4. Search results for "environmental monitoring sampling" in the AI support layer, showing a strong-match result (SOP-001) with three expandable panels: *Why this result?* (score breakdown by keyword index, title, section, and body text); *Related procedures* (adjacent documents surfaced from cross-reference mappings); and *Reasoning support note* (rule-based note flagging potential interpretive ambiguity without resolving it).

The AI support layer is organized around three design principles, each of which addresses a distinct component of interpretive burden identified in the pilot evaluation.

Explains why a result appeared. Each returned result includes a transparent account of its retrieval basis — which keywords matched, in which document fields, and at what weight. This addresses the opacity of ranked retrieval outputs, in which practitioners receive a result without understanding why it was returned. Making retrieval reasoning visible is not merely a usability enhancement; it is a condition for interpretive accountability. A practitioner who understands why a procedure was returned is better positioned to evaluate whether that basis is relevant to their task.

Surfaces related procedures. Each result includes adjacent documents surfaced from cross-reference mappings — procedures that are structurally connected to the returned result and that may be relevant to the same task. This addresses the scope ambiguity that participants identified in the pilot study: the difficulty of knowing whether a retrieved procedure covers the full range of a task, or whether adjacent procedures define its boundaries. Surfacing related procedures does not resolve this ambiguity; it makes the comparison space available for practitioner evaluation.

Keeps interpretive options open. Each result includes a rule-based reasoning support note that surfaces common ambiguities associated with the retrieved procedure — cases where similar tasks are governed by overlapping procedures, or where terminology in the procedure may apply to more than one process context. These notes do not recommend a compliance decision; they flag that multiple interpretations may be viable and that practitioner judgment is required to resolve them. The interface makes this constraint explicit: "this note keeps alternative interpretations visible. It does not replace reading or make compliance decisions."

The layer also addresses the no-result ambiguity identified in the pilot study through an explicit design principle: empty results are surfaced as signal rather than failure. When a query returns no results, the interface indicates that this is information about the archive structure — a possible gap in coverage or a terminology mismatch — rather than evidence of user error. This reframes retrieval failure as a structural finding, consistent with the broader argument that accessibility problems are properties of the archive rather than of the practitioner.

The AI support layer is governed throughout by the judgment-preservation constraint that applies to all computational support in GMP-regulated environments [1]: the layer expands the cues available to practitioners without producing outputs that function as quality decisions. It does not recommend procedures, evaluate compliance, or resolve interpretive ambiguity. Responsibility for reading, applying, and documenting procedural judgments remains entirely with the practitioner.

6 Discussion

Accessibility as a Design Condition on Interpretation

The central finding of this study is not simply that a retrieval tool reduces search time — though it does — but that access conditions shape the interpretive space within which practitioners act. When SOP retrieval is structurally difficult, practitioners do not simply search less efficiently; they act within a personally known subset of the procedural archive, consulting familiar procedures rather than searching for those that might be more applicable. In GMP environments, where SOPs are the formal basis for authorized action, retrieval constraints translate directly into constraints on what a practitioner can justify doing. The interpretive burden therefore begins not at the moment of reading a procedure but upstream, at the moment of deciding which procedures are reachable.

This finding extends Suchman's (1987) account of situated action. Applied to SOP use, the implication is that the formal existence of a procedure does not guarantee its practical force. What matters is what is accessible at the moment action is required. Our findings show the consequences of this when accessibility is structurally constrained: practitioners develop implicit repertoires bounded by what the archive makes findable, and these repertoires persist even when more applicable procedures exist elsewhere in the corpus. That Krockenberger et al.'s (2008) 18.4% colleague-consultation finding has persisted across decades of procedural system development is consistent with this interpretation: the structural conditions producing retrieval substitution have not been adequately addressed because they have not been treated as an interaction design problem.

Invisible Work Across Experience Levels

The onboarding finding illustrates the mechanism that Star and Strauss [12] describe as invisible work. Senior practitioners' informal SOP guidance labor is real, recurring, and consequential: it consumes experienced practitioners' time, creates informal dependencies between staff, and shapes how new practitioners develop their procedural knowledge. It is invisible in the specific sense that Star and Strauss identify: it does not appear in any quality metric, is not captured

by deviation reports or CAPA records, and is not attributed to the archive structure that produces it. The prototype partially surfaced this invisibility by making nomenclature disambiguation possible without colleague intervention — moving a category of informal coordination labor into the design layer, where it can be addressed structurally rather than managed interpersonally.

The broader implication is that accessibility failure redistributes labor asymmetrically. Practitioners who already know the system's naming conventions and archive logic can work around poor retrieval infrastructure; those who do not must either invest additional search labor or rely on human mediation. Accessibility is therefore not simply a property of documents. It is part of how epistemic access is distributed across a workforce.

Judgment-Preserving Design in Regulated Environments

The AI support layer introduced in Section 5 instantiates a design principle that Amershi et al. [1] formalize for human-AI interaction: AI systems should support human control, make their capabilities and limitations clear, and avoid producing outputs that pre-empt human judgment. In GMP environments, this principle takes on regulatory force: any system that produces outputs that function as compliance decisions introduces a traceability problem for the quality record itself. The AI support layer is designed to expand interpretive cues rather than resolve interpretive questions — making it a judgment-preserving intervention rather than a decision-support system in the conventional sense.

This distinction matters analytically. The design goal is not to remove ambiguity from procedure use but to make ambiguity legible enough that practitioners can compare possibilities without collapsing prematurely onto one because the archive made alternatives hard to reach. In this sense, the retrieval layer and the AI extension instantiate a shared design logic: both are comparison-preserving interventions that expand the set of alternatives a practitioner can hold in view, at different points in the interpretive sequence. Retrieval does this by expanding which procedures are reachable; the AI layer does this by keeping the comparison space open once procedures have been reached.

Design Implications for Regulated Document Environments

This study identifies a design space with no direct precedent in existing retrieval system research: external retrieval layers that operate over validated archives without modifying source files. Several design principles emerge. First, keyword-based access that decouples retrieval from archive naming conventions reduces the expert knowledge required to search effectively. Second, result cues that provide enough context to distinguish between adjacent procedures reduce the cost of navigating high-ambiguity clusters. Third, no-result transparency — making it clear whether a null result reflects a wording mismatch or a genuine archive gap — is a design requirement, not an enhancement; the AI support layer addresses this through the "empty results as signal" principle. Fourth, retrieval tools in regulated environments should be understood not only as efficiency infrastructure but as systems that shape what procedures remain visible and comparable at the moment of judgment.

Limitations and Future Work

Several limitations bound the interpretation of these findings. The synthetic corpus has less terminological and structural variation than a live archive would, which may have both underrepresented navigational difficulty and reduced result discriminability. Participant recruitment from a single team within a single facility introduces potential for inflated usability and acceptance ratings. The study lacks a comparative baseline, as no prior HCI tooling existed in this context. Think-aloud sessions were exploratory and session-based; they do not support claims about long-term adoption,

sustained behavior change, or downstream effects on interpretation quality or regulatory compliance. The AI support layer has not been empirically evaluated; its design principles are grounded in pilot findings and in established human-AI interaction guidelines, but their effectiveness in live deployment remains to be established.

The most important direction for future work is deployment with a live procedural archive. Future work should also examine the interpretive consequences of improved access: if practitioners reach a wider range of procedures, does this lead to more consistent interpretation, or does it introduce new forms of ambiguity by expanding the comparison set? The AI support layer's reasoning notes and related-procedure surfacing are designed to address this question in principle; empirical evaluation will be required to establish whether they do so in practice.

7 Conclusion

This paper introduced SOP accessibility as an upstream condition on interpretive practice in regulated manufacturing environments. When retrieval is structurally difficult, practitioners act within a personally known subset of the procedural archive — not because they lack compliance motivation, but because the archive does not make a broader range of procedures reachable. This constraint shapes what procedures remain visible, comparable, and defensible before any formal task error occurs and before responsibility is later assigned.

The prototype-based pilot study demonstrated that an external retrieval layer — operating over metadata without modifying validated source files — can reduce this upstream burden. Eighteen of twenty participants found the system intuitive, all twenty indicated willingness to reuse it in live archive conditions, and participants anticipated substantial reductions in the invisible onboarding labor that senior practitioners currently absorb informally. More importantly, improved access changed what practitioners felt able to consider and justify, indicating that retrieval design affects interpretive conditions rather than merely navigational speed.

Three contributions follow. First, SOP accessibility is introduced as a design problem specific to GMP-regulated environments, where the validated-archive constraint creates a design boundary with no precedent in existing retrieval research. Second, the external metadata layer is demonstrated as a viable strategy for reshaping the interpretive conditions of procedure use within compliance constraints. Third, the AI support layer is proposed as a judgment-preserving extension that addresses interpretive burden beyond retrieval by making retrieval reasoning transparent, surfacing related procedures, and keeping interpretive options open without substituting for practitioner judgment.

The wider implication is that document accessibility in regulated environments is not a usability problem to be solved once and set aside. It is part of how accountability conditions are produced in practice — determining which procedures practitioners encounter, which alternatives they can compare, and which authorized actions remain within reach. Future work that deploys these systems in live regulated environments will be essential to understanding whether improved access translates not only into faster retrieval but into more equitable, auditable, and accountable interpretive practice.

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